

Assignment 1 — Forecasting: Classical Methods and a Neural Network

Individual • 100 points • Released Wed, Sep 23 • Due Tue, Sep 29, 11:59 PM on Canvas

Submission instructions. Submit on Canvas. Responses may be typed or handwritten. Show your full problem-solving process; final answers alone will not receive full credit. You may use calculators and software tools (e.g., Excel) for your calculations. Combine all written answers into a single PDF; the Colab notebook is submitted as a second file (see Submission at the end).

What this assignment covers. Part 1 is the classical forecasting toolkit of Lecture 1, by hand: moving average, exponential smoothing, and double exponential smoothing, judged with MAD, MSD, and MAPE. Parts 2–5 turn the in-class Colab lab into graded work: the completed notebook, its scorecard, one controlled experiment, and a hand-worked theory part.

Materials. On Canvas: the Lecture 1 slides, the lab slides and Lab Guide, MFG510_NN_Forecasting_Colab.ipynb, and demand_monthly.csv.

Part 1 — Classical forecasting by hand (25 points)

Use the following three methods to forecast demand using the data in the table:

- (a) Moving average ($m = 2$)
- (b) Exponential smoothing ($\alpha = 0.2$), initialized with $F_1 = 8$
- (c) Double exponential smoothing ($\alpha = 0.2$, $\beta = 0.2$), initialized with $F_1 = 8$, $T_1 = 0$

Month	Demand
January	8
February	9
March	14
April	17
May	18
June	16

1. **Forecasts (15 pts).** Produce (1) one-step-ahead and (2) two-step-ahead forecasts for April–June with each method (a)–(c), using the formulas in the lecture slides. Show the intermediate values (each month's level F_t and trend T_t), not only the final forecasts.
 - A one-step-ahead forecast for month $t+1$ uses data up to and including month t (e.g., April uses data up to and including March). A two-step-ahead forecast for month $t+2$ uses data up to and including month t (e.g., April uses data up to and including February).

Notation (as in the lecture slides). F_t is the level estimate at the end of month t , and $f_{t+\tau}$ is the forecast made at the end of month t for month $t + \tau$. Moving average: $f_{t+\tau} = (A_t + A_{t-1})/2$ for $m = 2$. Exponential smoothing: $F_t = \alpha A_t + (1 - \alpha)F_{t-1}$ and $f_{t+\tau} = F_t$. Double exponential smoothing: $F_t = \alpha A_t + (1 - \alpha)(F_{t-1} + T_{t-1})$, $T_t = \beta(F_t - F_{t-1}) + (1 - \beta)T_{t-1}$, and $f_{t+\tau} = F_t + \tau T_t$. The initial values $F_1 = 8$ and $T_1 = 0$ are the level and trend at the end of January. Forecast error: $\epsilon_t = f_t - A_t$.

- 2. Accuracy (10 pts).** (1) Calculate MAD (mean absolute deviation), MSD (mean squared deviation), and MAPE (mean absolute percentage error) over April–June for each of the six forecast series (three methods $\times$ two horizons), and (2) compare the three methods on each metric: which method is best at each horizon, and does the ranking change from one-step-ahead to two-step-ahead? Explain in 3–5 sentences using the lecture ideas (lag, smoothing, trend).

Part 2 — Complete the notebook (10 points)

Fill every blank (the cells marked YOUR TURN) so that every CHECK cell prints ✓, and run the notebook top to bottom, including Cells 18–19. Then download it with the outputs included: File $\triangleright$ Download $\triangleright$ Download .ipynb.

- Graded on: all five CHECK cells pass (6 pts); all cells executed with outputs visible (4 pts). Syntax and coding style are not graded.

Part 3 — Scorecard and interpretation (15 points)

- 1. Scorecard table (5 pts).** Report MAD and MAPE on the 46 held-out months for: naive (copy last month), seasonal naive (copy the same month last year), the network trained from scratch (Cell 13), and baseline + network (Cell 18). One table, four rows.
- 2. Interpretation (4 pts).** One paragraph (≤ 150 words) in the style of Lecture 1, Section 2 (Evaluating demand forecasts) — e.g., "On the 46 held-out months, the network's MAPE is about ..., within ... of the seasonal naive and ... the naive error." State which forecast you would hand to planning and why.
- 3. Discussion questions (6 pts).** Answer the three questions from Cell 17 in 2–3 sentences each, in your own words: (a) why the seasonal naive is so strong on this series; (b) which method suffers from the test months sitting at a higher level than any training month, and why; (c) whether 182 training rows is a lot for ~5,800 weights.

Part 4 — One controlled experiment (30 points)

Cell 19 provides `train_and_score(...)`. In Cell 20, change exactly ONE argument from the lab configuration and run it. Choose one:

Option	Call	Lecture concept to use in your explanation
(a) smaller learning rate	<code>train_and_score(lr=0.001)</code>	learning rate / step size; convergence speed
(b) no month switches	<code>train_and_score(use_month=False)</code>	seasonal inputs; what the twelve lags already carry; rows vs. weights

Option	Call	Lecture concept to use in your explanation
(c) short window	<code>train_and_score(window=3)</code>	the sliding window; what information the inputs carry
(d) no early stopping	<code>train_and_score(early_stopping=False)</code>	overfitting; validation loss; early stopping

Report, in this order:

- **What you changed (2 pts).** One line: the argument, its lab value, and its new value.
- **Results (8 pts).** A two-column scorecard — lab configuration vs. your change — with MAD and MAPE for the network, plus the naive and seasonal-naive rows for reference; and the learning curve of your run (the figure Cell 20 prints).
- **Explanation (15 pts).** ≤ 200 words. Say whether the change helped or hurt, and explain the mechanism using the lecture concept in the table above. A correct explanation of a result that surprised you earns full credit; a vague "it got worse" does not.
- **Reliability check (5 pts).** Run your experiment a second time with `seed=1` (e.g. `train_and_score(lr=0.001, seed=1)`). Did the conclusion survive? One or two sentences.

Part 5 — Theory, by hand (20 points)

Show your work. Handwritten and scanned is fine. No code.

1. **Forward pass (8 pts).** A network has two inputs, one hidden layer of two ReLU units, and one output unit (a weighted sum plus a bias, no ReLU). The inputs are the last two quarters of demand scaled by 100: $x = (0.8, 1.4)$. Weights:

$$W(1) = \begin{bmatrix} 0.2 & -0.4 \\ 0.3 & 0.5 \end{bmatrix} \quad b(1) = (0.1, 0.2)$$

$$W(2) = \begin{bmatrix} 0.8 & 0.5 \end{bmatrix} \quad b(2) = 0.4$$

Compute the hidden pre-activations $z = W(1)x + b(1)$, then $h = \text{ReLU}(z)$, then the output $\hat{y} = W(2)h + b(2)$, and state the demand forecast in units. Which hidden unit switches off, and what does that mean in terms of regions of the input space?
2. **Pseudocode (8 pts).** Write pseudocode (≤ 12 lines) for ONE epoch of mini-batch gradient descent on the training table: inputs are the K rows (x_k, y_k), the current weights W , the learning rate η , and the batch size B . Use the loss $J(W) = \text{mean absolute error}$. Name where the forward pass, the gradient (backpropagation), and the weight update happen. You do not need to derive the gradient.
3. **Short answer (4 pts).** In 2–3 sentences: why must the test set be the most recent months rather than a random 20% of months? What would a random split make the test score measure instead?

Submission

- Two files on Canvas: Lastname_Firstname_A1.pdf (one PDF with Parts 1, 3, 4, and 5; Parts 1 and 5 may be scanned handwriting; ≤ 6 pages including figures) and Lastname_Firstname_A1.ipynb (the notebook of Part 2, with outputs).
- Individual work. You may discuss ideas with classmates; all calculations, code changes, numbers, and writing must be your own. Network numbers vary from run to run — that is expected; report what you obtained.
- Late policy and academic-integrity policy: as stated in the syllabus.

Rubric summary

Part	Points	Full credit looks like
1 Classical forecasting	25	Correct one- and two-step-ahead forecasts for all three methods with the intermediate levels and trends shown; correct MAD/MSD/MAPE for all six series; a comparison that names the best method per metric and horizon and explains why.
2 Notebook	10	All five CHECK cells ✓; every cell executed with outputs.
3 Scorecard & interpretation	15	Correct four-row table; paragraph in the language of Lecture 1, Section 2, with a recommendation; three discussion answers that use the lecture ideas (baseline assumptions, extrapolation, rows vs. weights).
4 Experiment	30	One argument changed; lab-vs-change scorecard and learning curve; a mechanism-level explanation using the named concept; a second-seed reliability check.
5 Theory	20	Correct z , h , $\hat{y}$ and forecast with the switched-off unit identified; pseudocode with forward pass, gradient, and update clearly placed; a precise answer on time-ordered testing.

Questions: office hours (ISTB12 355J) or Piazza.